

DEFENSE STANDARDIZATION PROGRAM

CASE STUDY

Body Armor Test Methods Standardization Program

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This case study describes a joint effort by the U.S. Army, the National Institute of Justice (NIJ), and the National Institute of Standards and Technology (NIST) to harmonize the test method standards and related standards applied to ballistic-resistant protective gear, such as helmets and vests. Each test method and related definitions in use at the start of this program had evolved to meet that community's (law enforcement or military) unique needs; as a result, standards and practices across the communities have been somewhat inconsistent. The technical insights and lessons learned from this multi-agency partnership can be carried over into standards efforts for other personal armor systems or other standards harmonization work.

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Defense Standardization Program Case Study

Body Armor Test Methods Standardization

Background

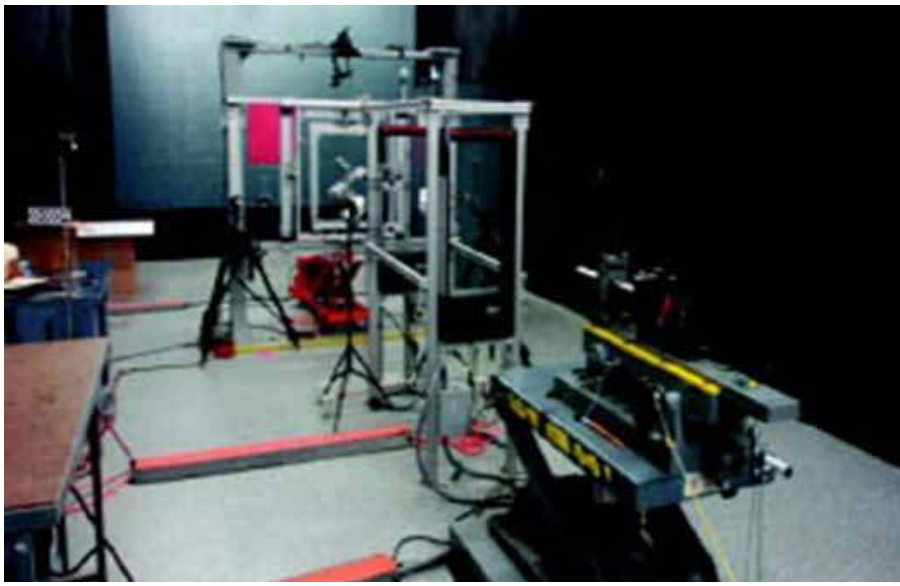
When an agency procures body armor to protect its personnel in the field, it needs to know that the armor will perform as expected and will meet requirements. The best way to provide that assurance is to have a robust testing program. To test personal body armor, it must be subjected to the attacks that it would be subjected to in the field; in the case of body armor for the Army and NIJ, the armor needs to be shot under controlled conditions. To conduct such a test, the lab needs a test range that includes the following:

- A way to consistently fire the projectile at known speed (velocity)
- Equipment that measures
  - the actual speed of the projectile;
  - conditions of the test, such as temperature, humidity, and distance the projectile traveled; and
  - the damage done to the model of a human body area—how much was the model behind the armor deformed or damaged?
- A method to mount the test item to receive the projectile, on a model of the human body part that is capable of recording damage (these are predominantly made of a specific modeling clay)
- The armor item being tested (e.g., protective vest, helmet)
- Additional items.

Figure 1 depicts these items in a photograph of a body armor test range.

Depending on the threat and the product being tested, different configurations of protective garments and armor plates are used.

Figure 1. A Body Armor Test Range at Aberdeen Testing Center (National Academy of Science, 2012)



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Problem

The U.S. Army and NIJ, with the support of NIST, have developed body armor standards and requirements in response to the needs of the communities they serve; each has responded to changes in threats as well as technology advances according to its own timetable and standards development processes. For example, changes in threats may require a new standard for performance required of body armor, while a technology change may affect both performance and the assessment/testing processes for body armor components. So, from the 1970s until recently, the Army was using its unique specifications and standards, while NIJ was using those contained in its standard, "Ballistic Resistance of Personal Body Armor Standard 0101", and NIST was providing support in areas such as initial test methods. The Defense community and the Justice community worked independently and used test specifications that were very similar—but they were not identical. Further, the different military services within DoD used different criteria for testing body armor.

Need for Harmonization

In the early 1970s, the law enforcement community needed a standard not only to define the physical requirements of body armor, but also to specify testing criteria to ensure that equipment met those requirements. The first standard for law enforcement body armor was published by the National Institute of Law Enforcement and Criminal Justice (NILECJ), the predecessor to NIJ, in March 1972, based on work conducted by the National Bureau of Standards (as NIST was known at that time). The NILECJ funded numerous projects, including some work involving DoD that correlated backface measurements (damage to the model) to blunt trauma and survivability data. The original NILECJ standard has been updated 5 times, with version 0101.06 published in 2008.

In DoD, the Army's Program Executive Office–Soldier (PEO Soldier) is responsible for body armor specification, procurement, and testing. In the 1970s–1990s, armor was built to Army specifications, which were very prescriptive. In the late 1990s, PEO Soldier worked with the Army's Natick Soldier Center to develop a performance-based purchase description (PD) for the next-generation Interceptor body armor. That PD included how PEO Soldier would test the products delivered, and it was continually improved with advances in technology made over the next 9 years. When Interceptor deliveries began in 2004, the Army did not have a lab with experience in the type of testing needed; PEO Soldier used the Army's Aberdeen Test Center (ATC) facilities and NIJ's first approved test lab, HP White. Inconsistencies existed within and between labs, and there was a need for faster fielding to troops in Afghanistan and Iraq. The General Accountability Office (GAO), the DoD Inspector General (DODIG), and the National Research Council (NRC) all wrote reports regarding needed improvements in DoD body armor testing. One recommendation was that DoD develop a test operations procedure for body armor ballistic inserts and ensure that the procedure was implemented DoD-wide. Another recommendation was for an indepen-

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dent assessment of First Article Testing data and an external peer review of the ATC facilities, instrumentation, and test protocols. In 2013, PEO Soldier started its collaboration with NIST and NIJ, and the ATC started participating in the collaboration soon after that.

Improving Standards

Throughout this time NIST was working with both the military and law enforcement users of body armor—the Department of Justice established a Law Enforcement Standards Laboratory (LESL) at the National Bureau of Standards (now known as NIST) in the early 1970s. The LESL's purpose was to recommend performance standards as voluntary aids for the selection of equipment by law enforcement agencies, and to develop standard test methods and procedures, so available armor products could be evaluated by those agencies. Since 2009, NIST laboratories have been working with the ATC to evaluate metrology techniques (e.g., using laser scanners) and analysis software. NIST even developed backface deformation artifacts to be used as a check standard, to ensure a lab's measurement system is performing properly.

Because of this history of working together, a collaboration between NIJ, the U.S. Army, and NIST to harmonize body armor standards was an obvious step toward improving those standards.

Approach

As mentioned above, there were studies by GAO, the DODIG, and the NRC; in a 2010 report, the NRC recommended that the community convene a nationally recognized group to establish a single national standard for handling and validating the clay used in testing.

In 2013, the U.S. Army, the National Institute of Justice, and the National Institute of Standards and Technology began a partnership to harmonize, to the greatest extent possible, the standards and their implementation for ballistic-resistant vests and helmets. The team members are working together to develop and revise standard test methods and practices.

ASTM International, a private-sector, American National Standards Institute–accredited standards developing organization (SDO), through its International E54 Committee on Homeland Security Applications, is the focal point in improving and validating methods, increasing consistency between test laboratories, and ultimately increasing confidence in ballistic-resistant equipment. Participants on the ASTM E54 Committee include representatives from federal agencies, law enforcement agencies, manufacturers, testing laboratories, and research organizations as well as body armor wearers.

The Dimensional Metrology Group in NIST's Physical Measurement Laboratory has completed a three-phase task for the Army ATC at Aberdeen Proving Ground, MD. All three phases were related to the specification and verification of the body armor plates worn by U.S. soldiers in battle. As these armor plates are critical to the safety of our soldiers, strict adherence to measurement standards can literally mean the difference between life and death. NIST delivered calibrated back-

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face deformation check standards for ballistics testing of the body armor, calibrated surrogate body armor check standards for the armor's dimensions and geometry, and also provided technical improvements to geometric dimension and tolerance drawings (essentially blueprints) of the armor's dimensions.

Use and Development of Consensus Standards

The 1996 National Technology Transfer and Advancement Act directs that all federal agencies use technical standards developed by voluntary consensus standards bodies, where possible, as opposed to using government-unique standards. Further guidance is provided by Office of Management and Budget Circular A-119 on implementation of the act and establishes federal policy on the development and use of voluntary consensus standards. Many SDOs in the United States operate under guidelines from the American National Standards Institute, the official U.S. representative to the International Organization for Standardization.

Federal agencies working with SDOs to develop and implement voluntary consensus standards can receive such benefits as the following:

- Saving the costs of developing government-unique standards, while participating in the development of standards that serve national needs
- Simplifying the acquisition of goods and services procured by the government, promoting greater efficiency and economic competition
- Decreasing the cost of goods and services procured by the government
- Promoting efficiency and economic competition and encouraging long-term growth for U.S. enterprises
- Furthering the government policy of reliance upon the private sector to supply goods and services needed by the federal government, including standards.

The next section, "Benefits of This Approach," discusses the specific benefits of harmonizing body armor standards via the private-sector voluntary consensus development process.

The Army, NIJ, and NIST partnered in 2013 with the aim of harmonizing both the standards and their implementation for ballistic-resistant vests. NIST is a critical partner, because it is the point of coordination for all body armor standards work, on behalf of NIJ and the Army. The partners determined that ASTM International's E54 Committee on Homeland Security Applications was the group to work with to develop and maintain standards and other guidance for ballistic-resistant equipment. As mentioned earlier, the committee would focus on test methods and practices for the purpose of improving and validating methods, increasing consistency between test laboratories, and ultimately increasing confidence in ballistic-resistant equipment.

First, the Army NIJ and NIST partners guided the analysis of the corpus of documents used by NIJ and the Army, to identify those that contained test methods and practices. Then they determined which of those would be candidates for harmonization. For example, some documents were

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properly unique to either NIJ or the Army because of unique threats and test item or performance requirements. Other documents could be harmonized because they dealt with standards and practices shared by both—for example, the terminology used, configuring a test range, the specification for the clay used, and the procedure for building it up for use behind the armor plate.

ASTM International provides the forum and infrastructure to enable all interested stakeholders to collaborate. In addition to the Army, NIJ, and NIST, the larger committee includes people representing other federal agencies, law enforcement agencies, manufacturers, testing laboratories, research organizations, and body armor wearers. This holistic approach involving all interested stakeholders not only produces high-quality standards, but also creates a community that works well together for the improved protection of law enforcement officers and soldiers.

Benefits of This Approach

Harmonizing body armor standards via the consensus development process is yielding many benefits for stakeholders:

1. The community's development of standard terminology enables effective communication, shared understanding, and a common approach to talking about issues. Before having this standard terminology, body armor terms had multiple and different meanings depending on who was using the term, leading to possible confusion and impaired communications.

2. Working through an SDO provides openness, transparency, and a forum for stakeholder discussions, maintaining a balance of interests and due process. The SDO and its processes are trusted, which leads to greater buy-in from the community. This is the first time that all stakeholders, including manufacturers, have been invited to the table to participate in the development of standards for NIJ and the Army. As a result, federal agencies are now able to tap into the collective knowledge of all participating stakeholders, end users, manufacturers, and test laboratories that benefit from having a stronger voice in the standard development process.

3. There is increased consistency of testing with decreased testing costs. This is a result of reducing the number of unique standards. For example, both NIJ and the Army need the preparation and verification of the clay backing placed behind armor during ballistic testing; however, they have been using different methods, equipment, and requirements for verifying that the clay is ready for use. Moving forward, both agencies will use the same ASTM method, where they can be continually reviewed and

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updated according to the state of the art for body armor testing and the emerging needs of end users. ASTM standards follow a 5-year cycle and may be revised as often as the stakeholders deem necessary.

4. An unanticipated benefit is that the community of stakeholders has carried its experience with body armor standards harmonization into the development of an ASTM test method for ballistic helmets that is applicable to both law enforcement and soldiers. It is likely that this harmonization will continue to propagate into standards development in other fields.

5. PEO Soldier has seen big gains overall. There has been a reduction in complaints from contractors, a quicker response to soldiers' needs, and improved consistency in the product. Products have improved and armor is fielded faster. The faster delivery of consistent armor products means that more soldiers' lives are saved.

Current Status and Future Plans

The first two in a family of about 15 standards have been published:

- ASTM E3004-15, "Standard Specification for Preparation and Verification of Clay Blocks Used in Ballistic-Resistance Testing of Torso Body Armor" (ASTM International, 2015)
- ASTM E3005-15, "Standard Terminology for Body Armor" (ASTM International, 2015).

Another dozen items identified for harmonization are currently under development as ASTM work items, and several will be maintained independently (not harmonized) by the Army or by NIJ, as shown in Figure 2.

Each of these items is in a different stage of development, but the overall goal is to publish all harmonized ASTM body armor standards by the end of 2017. Not all documents will be harmonized; those on the left in Figure 2 are better maintained as organization-unique standards.

As each of the harmonized standards is published, NIST will assist the Army and NIJ with integration of the ASTM standards into the documents they will be independently maintaining. Once the standards have been integrated, the NIST National Voluntary Laboratory Accreditation Program will begin to accredit test laboratories to the standards. The harmonization process is already benefiting the body armor community, and the benefits will continue to be realized for years to come.

Lessons Learned

Many lessons have been learned; the first was that there were multiple parallel efforts to develop standards that covered very similar topics. This revealed the need to work together to develop joint standards. Next, the organizations learned how the SDOs' standard development process could accommodate different organizations, agencies, and technical areas.

SDOs establish groups of subject matter experts (SMEs) on a particular topic or practice. Participation in a group is open to members who have both an interest and expertise in that topic/

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Figure 2. Items That Will Remain Unique and Items That Will Be Harmonized

Maintained NIJ and U.S. Army Unique Documents

- Test Threats
- Test Item Requirements
- Performance Requirements

Standards, Test Methods and Procedures, and Practices Appropriate for Harmonization

- Standard Terminology for Body Armor
- Clay Block Backing Material Specification
- Test Range Configuration
- Hard Armor Conditioning Procedures
- Backface Deformation Measurement
- Procedure for Clay Build-Up behind Nonplanar Soft Armor for Females
- Procedure for Clay Build-Up behind Hard Armor Plates
- VSO Ballistic Limit Test
- Perforation-Backface Deformation Test
- Projectile Velocity Measurement
- Labeling Specification
- Workmanship Requirements
- Carrier Specification



practice and who have time to volunteer for the standard effort. A standard's development typically begins with group discussions to determine what is in or out of bounds of the standard being developed.

As part of that boundary discussion, a group must consider how "atomic" the topic is—can it be broken into constituent parts? Then, can standards be written at the level of each part—for example, can test methods be separated from performance requirements and be treated separately from threats? If so, there are advantages to these separate standards:

- An organization can more easily adopt those parts that apply to its overall process.

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- An individual standard can be updated in response to changes in its environment (technological change, etc.).
- Updating a given standard is easier when it treats one standalone topic—the standard is smaller and more focused.

The groups will then draft the standards based on current practices—discussing what is considered "best practice" in their respective organizations. Over time, this results in a distillation of the best practices, documented in a way that they can be tailored to each unique organization; this is the draft standard. It will continue to be developed and revised until the group decides it has a final draft.

This process is, by its nature, deliberate and therefore not quick; it may take months to more than a year to develop a standard that is ready for release. Depending on the SDO, the standard may be approved by votes from members of other SME groups who are knowledgeable about the topic (this is part of the consensus process) or released after an effort to gather and incorporate comments from others. No matter the mechanism for consensus, there is a way to have review and comment from SMEs outside the standard developer group before release.

The organizations have also learned that the bigger challenges in this work are not technical but are relational. This is not meant to imply that there are no technical issues, because there are complex issues that require research and testing to address. But it is more challenging bringing together people whose organizations are competitors in the marketplace—while having them set that aside and contribute their intellectual capital for the good of the community. Coming to consensus requires that individuals learn to listen to and value different perspectives, overcoming the temptation to argue against or vote down contrary perspectives. This is often challenging for experts who have vast knowledge and experience. Over many months of meetings and discussions, the individuals involved began to grasp this concept and have evolved into a team in which the members value each other's perspectives.

A final lesson learned is that, by listening with an open mind to how other organizations manage similar challenges, an organization can improve its own internal processes. Often participants in the standard development group bring back fresh perspectives not only on the group topic, but on how related practices can be improved. This is a benefit, but also an example of how competitor company representatives carefully discuss their practices without giving away any proprietary information, while still contributing to a rich dialogue and discussion. Over time, a standards group will evolve into a gathering point for some of the best, tested, most knowledgeable information on the topic.

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